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How Not to Measure Whether a Judge Knows It Is Wrong

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Abstract

Asking whether an LLM judge’s internal state predicts its own errors is a type-2 (correctness) ROC problem, and the standard way of computing it is wrong in five separable ways. We measure each. First, because a judge is correct exactly when its verdict matches the label, any feature that merely separates positive from negative items scores well while carrying no self-knowledge; identity decodability runs 0.599–0.899 on our benchmark irrespective of which model judges. Second, and not removed by conditioning on the label, within a label stratum *correct* means *the judge said this stratum*, so a representation decoding the judge’s own verdict is a perfect within-stratum predictor — and the last-token hidden state decodes it at 1.000. What blocks this is the pooled fit, not the metric, and only when the strata are balanced. Third, concatenating a hidden-state block onto a small baseline under a shared penalty does not measure increment: a block of 4096 *pure-noise* columns costs a strong baseline 0.315 AUROC, and the damage grows with baseline strength. Fourth, the field-default paired DeLong test understates the standard error by $2.17\times$ on pairwise judge data, which is clustered by construction. Fifth, the reference point is not 0.5: a first-order signal-detection observer with no metacognition whatever reaches 0.765 at accuracy 0.802, so type-2 AUROC scored against 0.5 credits first-order accuracy as self-knowledge. We give a corrected estimator for each, with a regression test apiece, and report a demonstration panel of confidence-margin, activation-probe and attention-graph spectral features under it. The one result that survives every correction unchanged is an equivalence: spectral features add nothing beyond a linear activation probe, with deltas in -0.012 – $+0.000$ at standard errors of 0.000 – 0.011 .

1 Introduction

LLM judges are used to score model outputs at a scale that forbids human review, so knowing when a judge is wrong is worth something. A natural proposal is to read it off the judge itself: train a probe on the judge’s internal state to predict whether its verdict was correct, and report the AUROC. The construction is standard, it is cheap, and it produces numbers between 0.6 and 0.9 with reassuring regularity.

Those numbers are hard to interpret, for a reason visible in the target variable. Correctness is defined as $\text{is_correct} = (\text{pred} = \text{gt})$, a comparison of two things, and a probe can score well by knowing about either one without knowing anything about the judge’s own reliability. This paper takes that observation seriously and follows it through the estimator.

We find five separable failures. Two are leakage channels: one through the label, one through the judge’s own verdict, and they require different fixes (Section 5). One is in the ladder that is supposed to measure what internals *add* to behaviour, which under a shared penalty measures something else entirely and does so in a direction that makes internals look *worse* than they are (Section 6). One is in the inference, where the field-default paired test assumes an independence that pairwise judge benchmarks violate by construction (Section 7). And one is in the reference point, where 0.5 implies a comparison to a judge that cannot decide anything, rather than to a judge that decides as well as this one but knows nothing about how it decided (Section 4).

Four of the five inflate the reported number and one deflates it, which is worth stating plainly: this is not an argument that internal signals are weaker than claimed, it is an argument that the published quantity is not the one anybody wants.

Contributions. (i) We identify the type-2 estimand for judge metacognition and give it two floors rather than one: a first-order signal-detection floor, which is the Monte-Carlo form of $\text{meta-}d'/d'$, and a verdict-leakage floor arising from incomplete sign cancellation at unbalanced strata. (ii) We give the two-dimensional leakage taxonomy — item identity and self-verdict — and show that the second is blocked by the pooled fit rather than by the conditional metric, making it

fragile to a refactor that looks innocuous. (iii) We show that the usual nested-model ladder does not measure increment at realistic block width, quantify the damage with a pure-noise block, and give a monotone offset estimator. (iv) We quantify the clustering penalty for paired DeLong on pairwise judge data. (v) We report a demonstration panel under the corrected estimator, and one equivalence result that survives all five corrections. (vi) We report our own analytic errors grouped by cause (Section 11), including two that were about to enter a write-up as findings.

2 Related work

Metacognition and the type-2 ROC. Distinguishing an observer’s first-order sensitivity from its insight into its own performance is the subject of a long methodological literature. $\text{Meta-}d'$ (Maniscalco and Lau, 2012) expresses metacognitive sensitivity in the units of first-order d' , and the ratio $\text{meta-}d'/d'$ — the M-ratio — is 1.0 when an observer has no insight beyond what its first-order performance already implies (Fleming and Lau, 2014). The essential point, which the probing literature on LLMs has not imported, is that a type-2 ROC computed against 0.5 is uninterpretable: a more accurate first-order observer scores higher on it without any metacognition at all. Section 4 is a Monte-Carlo M-ratio for judges.

Probing and its controls. That a probe’s score reflects the probe rather than the representation is the concern of Hewitt and Liang (2019), answered with control tasks; the follow-on literature pursues it in opposite directions — Voita and Titov (2020) recasting probing as compression, Pimentel et al. (2020) arguing from mutual information that one should choose the *highest*-performing probe rather than the simplest — and through whether the model *uses* what a probe can read (Ravichander et al., 2021; Elazar et al., 2021), synthesised by Belinkov (2022). Our positive control (Section 6.1) is in that tradition. The failure in Section 6 is different in kind and, as far as we know, undocumented: it is not about probe capacity but about what happens to a *nested* comparison when the two models being compared have very different column counts under one penalty.

LLM judges. Judge models are evaluated on benchmarks built for the purpose (Zheng et al., 2023; Tan et al., 2025; Zeng et al., 2024), whose known pathologies — position bias, verbosity bias and self-enhancement (Zheng et al., 2023) — motivate the confound register we carry alongside this work. The two free-text banks we treat as primary were chosen for complementary label semantics: Tan et al. (2025) labels responses by *objective* correctness rather than by human preference, and Zeng et al. (2024) is adversarial by construction, its 419 curated pairs pitting an instruction-violating response that is superficially attractive against a compliant one. Our judging setup is logprob readout, which is what reward models and constrained-decoding judge pipelines actually do at scale.

Selective prediction. The practical form of the question is abstention: given a signal, what precision is available at a fixed coverage (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017). We report coverage-at-fixed-risk rather than ΔAUROC for the applied claim, because a 0.02 AUROC difference has no operational meaning and a coverage change does.

3 Setup

Judging. A judge scores an item in one forward pass; the verdict is read from the next-token logprobs of two verdict tokens at the position following `Answer:`. The *margin* is the signed difference of those two logprobs, so the predicted verdict is a deterministic function of $\text{sign}(\text{margin})$ by construction — a fact that matters in Sections 4 and 5. Three formats: `mcq` and `single` (Yes/No) and `pairwise` (A/B, presented in *both* orders). Prompts are raw and identical across the panel; a chat-template ablation moves a competent judge by at most 1.5 points.

Banks. Five banks (MMLU, MMLU-Pro, JudgeBench, LLMBench, RewardBench 2), balanced 50/50 by construction, seeded, and byte-identical across judges. Cross-validation groups are derived from normalised question *text*, not identifiers: JudgeBench reuses 92 questions across its two splits, and grouping on identifiers would place the same question in train and test.

Features. Three families in increasing cost. **Margin:** the signed logprob margin and its absolute value. **Activation:** the last-token hidden state at half and full depth. **Spectral:** attention-graph diagnostics — Fiedler value, high-frequency energy ratio, smoothness, spectral entropy, plus a Fiedler-velocity summary — computed on the symmetric normalised Laplacian over a task-token subgraph, summarised as mean, slope and normalised argmax over the last third of layers. The summary is fixed *a priori*: three numbers per metric, no post-hoc layer selection. Precision is `bfloat16` uniformly (Section 9).

The ladder. Rungs are M_n (nuisance only: length, task-start index, subject), M_1 (margin and verdict), M_{1n} (+ nuisance), M_{1nd} (+ leave-one-model-out peer difficulty), M_2 (+ spectral), M_3 (+ activations), M_4 (everything). The baseline every internal claim must clear is M_{1nd} where at least two judges scored the same items.

4 The estimand and its two floors

4.1 Neither floor is 0.5

Write $y = \text{is_correct}$ and let the strata be the ground-truth verdict. The headline metric is the *conditional* AUROC: the probability that a correct item outscores an incorrect one *drawn from the same stratum*. Within a stratum the item’s label is constant, which removes the first leakage channel (Section 5.1).

It does not follow that 0.5 is the reference. Consider a first-order signal-detection observer: one latent decision variable s , verdict = $\text{sign}(s - c)$, margin = $s - c$, and nothing whatever that knows about its own errors. Its discriminability d' drives its accuracy, and it drives the informativeness of its margin at the same time. Simulating this observer and pushing it through the identical features, estimator and metric:

Table 1: A judge with *no* metacognition, at three discriminabilities. Type-2 AUROC scored against 0.5 would read the third row as strong self-knowledge.

	accuracy	M_1 conditional AUROC
$d' \approx 0$	0.505	0.466
$d' = 0.9$	0.802	0.765
$d' = 1.4$	0.943	0.815

This is the M-ratio of Maniscalco and Lau (2012) in Monte-Carlo form. We estimate each judge’s own d' and criterion from its hit and false-alarm rates, simulate the first-order observer at those values, and report every rung against *that* null rather than against 0.5. Parameter uncertainty is propagated: d' and c are re-estimated on each bootstrap resample, because simulating at the point estimate gives a null that is too narrow and makes “above null” anticonservative. Each rung receives a percentile against the null distribution, and those percentiles enter the same Benjamini–Hochberg family as every other contrast.

4.2 The verdict-leakage floor

The second floor is specific to this target. Within the stratum $\text{gt} = A$, an item is correct if and only if the judge said A . So a feature decoding only `pred` is a perfect within-stratum predictor. What prevents a pooled model from selecting it is sign cancellation: such a feature must score 1.0 in one stratum and 0.0 in the other.

Cancellation is exact only when the strata are balanced, and judges are biased — verdict bias runs from 0.14 to 0.64 across our pilot panel. We therefore measure the residual directly: a synthetic feature that decodes the judge’s verdict and nothing else, pushed through the full pipeline at each slice’s own observed imbalance. The conditional AUROC it reaches is that slice’s leakage floor, and it differs from the SDT floor. A rung must clear both.

5 Two leakage channels

5.1 Item identity

Because $y = (\text{pred} = \text{gt})$ and every judge has some verdict bias, y is largely determined by `gt` alone. Any feature separating a positive item from a negative one therefore scores without carrying a bit about self-knowledge, and spectral

and activation features do so trivially: the two prompts contain different text. We report `identity_decodability` — how well the same features predict `gt` — as a standing control. It runs 0.599–0.899 across our slices and, importantly, is *flat across judges of very different competence*: it is a property of the benchmark, not of the model being measured, so it contaminates every result reported on that benchmark and cannot be escaped by choosing a better judge. Conditioning on `gt` removes it.

5.2 Self-verdict

The twin channel is not removed by conditioning, because the quantity decoded is `pred`, not `gt` (Section 4.2). The last-token hidden state decodes `pred` essentially perfectly — it is the state the verdict logit is read from — and we report `verdict_decodability` beside the identity control for exactly that reason.

The protection is the *pooled fit*, which is a property of how the model is estimated rather than of the metric. Fitting within stratum — an innocuous-looking refactor — removes it and yields a near-perfect, entirely meaningless result. On planted data where a feature decodes `pred` and nothing else, the pooled fit lands near 0.5 and a within-stratum fit exceeds 0.95. We encode this as a regression test rather than a comment.

Leakage into type-2 AUROC is therefore two-dimensional: item identity, removed by conditioning, and self-verdict, removed only by pooled estimation at balanced strata. Probes controlling for neither are exposed to both.

6 The ladder does not measure addition

A nested comparison is supposed to answer: what does this block add to what we already had? Fitting it by concatenating the block onto the baseline columns under one shared penalty does not answer that question when the two differ in width by three orders of magnitude. A `hidden_size`-wide activation block against a six-column baseline yields approximately “activations”, not “baseline + activations”, and the block must reconstruct the margin from scratch. When it cannot, the rung scores *below* the baseline it was meant to extend.

Table 2: Planted validation. The added block is **pure noise**; the correct delta is zero at every width. Concatenation destroys real signal, and destroys more of it when there is more to destroy.

baseline	added block	concat	offset
0.882	32 noise cols	−0.029	±0.000
0.882	4096 noise cols	− 0.315	±0.000
0.745	4096 noise cols	−0.158	±0.000

The signature is diagnostic: the delta correlates negatively with baseline strength, so the better the baseline, the more the rung appears to lose. We observed exactly that on real data before identifying the cause.

The offset estimator. Fit the baseline first, carry its out-of-fold logit as a fixed offset, and let the internal block fit only the working residual under ridge shrinkage: $\text{score} = z_{\text{baseline}} + f(X_{\text{internal}})$. If the block is uninformative, f shrinks to zero and the score reduces to the baseline, so addition is monotone by construction and a negative delta means overfitting rather than destruction of information already in hand. The offset for training rows comes from an *inner* cross-validation inside the training fold; using the outer out-of-fold logit would leak the test fold into the baseline the second stage sees.

Two estimands, reported side by side. The offset rung answers “what does this block add to what we already had”. A block-only rung answers “what is achievable from this block alone”. The gap between them is redundancy. Both belong in the table: the offset alone reads as self-handicapping, and the block-only alone hides that the information was already present.

6.1 Absence versus redundancy

A tight equivalence between M_4 and M_3 has three explanations the delta cannot separate: spectral carries nothing; spectral carries what activations already carry; or the penalty shrank the spectral block to zero. A fourth is not

hypothetical (Section 9): the features never arrived. We separate them with a block-only rung, the fitted coefficient norm on the spectral block against the activation block, and a positive control that recovers a target known to be encoded at matched n . On planted data the two worlds give indistinguishable deltas (-0.0035 and -0.0032) and block-only rungs of 0.504 versus 0.876.

7 Inference

Clustering. The pos/neg partners of one question and the two orders of one pairwise item share nearly all their text and live in one cross-validation group. Paired DeLong assumes independent items. On planted data with one item per group its standard error matches the grouped bootstrap exactly (0.0020 both); with four clustered rows per question it understates the standard error by $2.17\times$. Since pairwise judge benchmarks are clustered by construction, adopting DeLong as the primary test manufactures significance. We use a grouped paired bootstrap, which is already paired — both metrics are recomputed on identical resamples — and report DeLong only as a labelled cross-check whose ratio to the bootstrap measures how much width is clustering.

Equivalence. A non-significant contrast is not a finding unless it says what it rules out. Every contrast carries a two-one-sided-test verdict against a pre-specified band of 0.02 conditional AUROC, read off the same bootstrap so the grouping is respected.

Power. The width of each bootstrap CI is the standard error at the pilot's n , so the pilot prices its own minimum detectable effect. At 200 items per slice and a median standard error of 0.006, the detectable effect is 0.015 uncorrected and 0.024 under Benjamini–Hochberg over 66 contrasts. Against a plausible target of 0.04, a pilot null is uninformative, and detecting it requires roughly 72 items per slice. Two consequences shape the design: buy items rather than judges, since each judge multiplies the contrast family without adding per-test power; and pre-specify 3 primary tests, with everything else exploratory and unstarred.

8 Results

[Pending the regeneration pass. `fill_numbers.py --verify` currently exits non-zero on 21 macros, all of which are downstream of it. This section is written to be filled, not to be guessed.]

The panel: ?? judges spanning scale, ?? items per slice, all five banks entering a pooled hierarchical model as a random effect rather than being tested separately. Both floors are reported per slice, alongside the per-stratum support counts, since a conditional AUROC resting on a handful of minority-class items looks exactly like one resting on hundreds.

Anchor: spectral adds nothing beyond activations. This is the one result unaffected by Sections 4 to 6: M_4 and M_3 carry the same baseline and the same verdict leakage, so their difference is clean under all of them. Across 6 estimable slices the delta lies in -0.012–+0.000 at standard errors of 0.000–0.011, which is TOST-equivalent against the 0.02 band. The attention-graph family is measurably redundant with a linear probe on the last-token hidden state, reported as a powered equivalence rather than as a failure to reject.

Controls. Permutation nulls, refitting the whole pipeline with labels shuffled within stratum, land in –, so there is no leakage in the cross-validation itself.

9 Numerical and provenance hygiene

Two failures in this class nearly entered our own write-up as findings, and both are silent.

Precision. Under `float16`, one judge overflowed to `inf` inside attention. Its spectral analysis failed on 100% of items — caught per item by design, so nothing crashed and no summary looked wrong — *and* its verdicts were corrupted, presenting it as a fully degenerate judge (50.0% accuracy, 100.0% verdict bias) where under `bfloat16` it is competent (73.5%). The two precisions agree on 36.5% of its verdicts. A numerical bug was on its way into a write-up as a finding about model scale. Precision must be uniform across a panel, since neither spectral metrics nor logits are comparable across dtypes. We have promoted the check into the analysis library itself, which now raises on the first layer of non-finite attention.

Versions. The pilot ran a different version of the spectral library than the environment later held, and nothing in the results files recorded it. Comparing rows computed by different library versions is the same class of error as mixing dtypes. We verified that the change on the code path in use shifts every metric by at most 0.02% and pinned the version.

10 Limitations

Named hazards we did not measure, each with the measurement we would propose.

Chain-of-thought. Our judges are single-token logprob readouts, which is what production reward pipelines do but is not how a judge is always deployed. The signal-detection null transfers unchanged — it needs only a two-way argmax over a scalar margin at the readout token — but generated chain length correlates with item difficulty and would need to enter the baseline as a nuisance covariate, exactly as prompt length does. We predict internals add *less* under chain-of-thought, because the model can externalise uncertainty into tokens the readout state has already absorbed.

Open-weight panel. Internal access requires open weights, so the judges most used in practice are absent.

single-format label noise. On preference datasets the “is this response correct” framing treats a relative preference as absolute truth, capping achievable accuracy. Pairwise is our primary format for this reason, and is additionally immune to the length confound by counterbalancing.

The plant-indexed recovery curve did not earn its keep on our own measurement. We index the curve on planted strength rather than on a recovered block-only AUROC because indexing on an estimated quantity is ill-posed in principle: the recovered value is biased by the same shrinkage that suppresses the delta. At the width and sample size we measured (1536 columns, $n = 200$) that bias turns out to be negligible — block-only is flat in n — so on *this* measurement the redesign’s practical benefit came from pairing the cells and seeding them, not from the bias correction that motivated it. We report both, and continue to recommend the plant-indexed form, because the bias is a function of width and sample size and there is no reason to expect it to stay negligible elsewhere.

Fixed spectral summarisation. Three numbers per metric, chosen *a priori* to avoid post-hoc layer selection. If the signal is layer-local, this design cannot see it, and a null would be about the summarisation rather than the representation. One pre-registered contrast on the full layerwise vector addresses this at the cost of a single test.

11 Corrections

We report our own errors grouped by cause, because the discipline this paper argues for is worthless if applied only to others.

Two artefacts that survived multiple-comparison correction. A judge answering one way almost always has near-zero within-stratum outcome variance; its conditional AUROC was estimated from a stratum holding a single minority-class item, and a bootstrap resampling that same item returned a *narrow* interval around an arbitrary value. Two such contrasts passed Benjamini–Hochberg and were briefly reported as discoveries. Fixed in three layers: a minimum class count per stratum, contrast suppression for degenerate slices, and exclusion of unreliable slices from the correction family.

A validation that could not detect the failure it was for. Our first planted test of the ladder used a 32-column block and reported it clean. At that width the pathology of Section 6 is 0.029 and invisible; at realistic width it is 0.315.

A trend we nearly reported. The margin’s conditional AUROC rose with judge competence across our panel, which reads as self-knowledge emerging with capability. Simulating the first-order observer of Section 4 reproduces the entire trend, and our judges sit at or below it. The finding was arithmetic.

Use of large language models

We disclose LLM involvement in this work in the terms the ICLR 2027 policy requires, and in more detail than the minimum, because the paper’s subject makes the question directly relevant.

As a methodological adversary. A substantial part of this paper originated in an extended adversarial exchange with a large language model (Anthropic Claude, Opus 5, accessed through Claude Code between July and September 2026). The model was given the repository and asked to attack the estimator rather than to defend it. Several of the findings reported here were first raised in that exchange and then verified by us in code:

- the observation that the ladder’s deltas correlated negatively with baseline strength, which led to the concat-swamping result of Section 6;
- the request for a matched signal-detection null, which produced Section 4 and retracted a behavioural trend we had been prepared to report as a finding;
- the identification of the self-verdict leakage channel and the observation that the pooled fit, not the conditional metric, is what blocks it (Section 5.2);
- the demand for a minimum-detectable-effect calculation before further compute (Section 7), which established that our pilot null was uninformative.

Every one of these was independently checked: each is reproduced by a planted-data test in the released test suite, and each is stated in the paper with the measurement rather than the argument. Where the model’s premises were wrong we say so in Section 11, and two of its recommendations were tested and rejected on evidence — notably the proposal to replace the grouped bootstrap with paired DeLong, which we show would have understated our standard errors by $2.17\times$.

As an implementer. The same model wrote or substantially edited code in the released repository, including the offset ladder estimator, the two null references, the leakage controls, the power stage and the regression tests. All of it is covered by the test suite and by the build checks described below; none of it was accepted without a test that fails when the behaviour regresses.

As a search tool over the literature, with human verification. Candidate references were located using LLM-assisted search. Every citation was then opened and checked by a human author against the claim the sentence makes, and the record of that check — what each paper establishes, and which sentence leans on it — is kept as structured comments in the released `refs.bib`. Two mismatches were caught this way and fixed in the *sentence* rather than the citation: a pair of probing papers that reach opposite prescriptions had been cited as agreeing, and a reward-model benchmark had been cited behind a later, differently-named dataset it does not describe. A build check refuses to compile the paper if any cited key lacks a resolvable DOI or arXiv identifier.

What was not delegated. The research question, the experimental design, the choice of banks and panel, the preregistered primary tests, and the decision about what may be claimed are ours. So is responsibility for everything in this paper, including anything an LLM produced.

Reproducibility of the disclosure itself. Because a claim about how a paper was written is as checkable as any other, the repository carries the full provenance: every result file records the git commit, the package versions — including the exact VCS commit of the spectral library, not merely its version string — the configuration and a digest of the bank that was scored. Every number printed in this paper is a macro resolved from a results file, and the build fails on an unresolved one rather than printing a stale value.

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